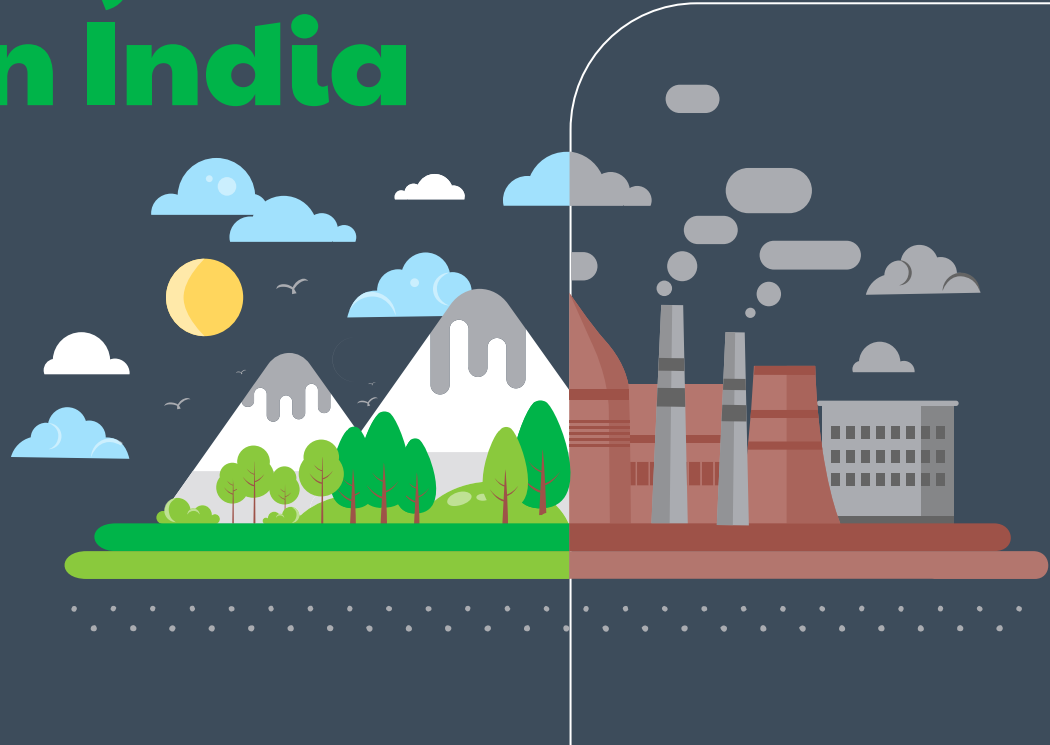


Carbon Capture, Utilization, and Storage in India

A technological alternative
to decarbonise Indian
industries

Vaibhav Agarwal
Abhinav Saxena



Objectives

- To introduce the problem of rising industrial emissions and the distribution of these emissions
- Reviewing CCUS as a technological solution to decarbonise Industries
- Identifying the environmental, social, and economic risks attached to CCUS, especially in the Indian context.

Hypothesis

Carbon Capture, Storage, and Utilization (CCUS) can be a potential technological alternative to decarbonise Indian industries.



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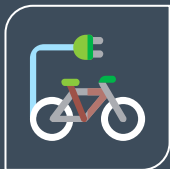
01

**Industrial
Emissions**



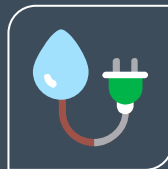
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**Introduction to
CCUS**



03

CCUS in India



04

**Results and
Discussions**



01

Industrial Emissions

An introduction to the problem.



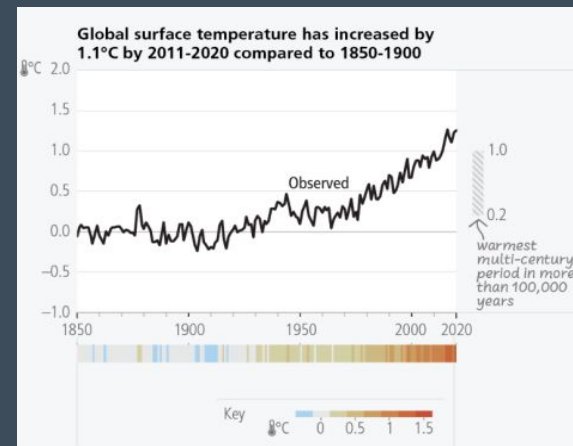
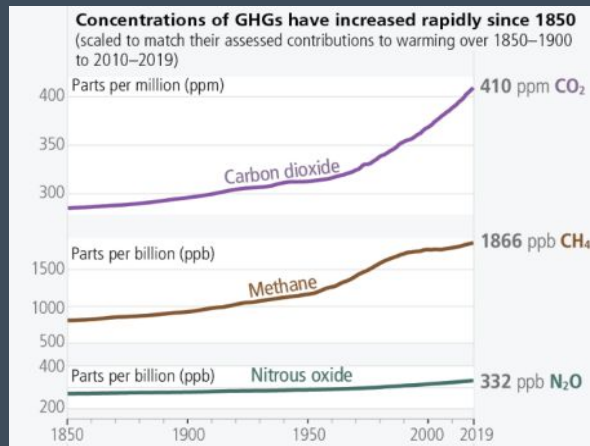
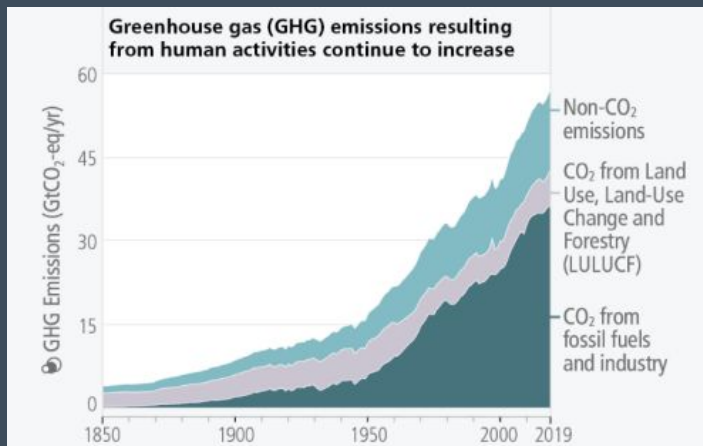
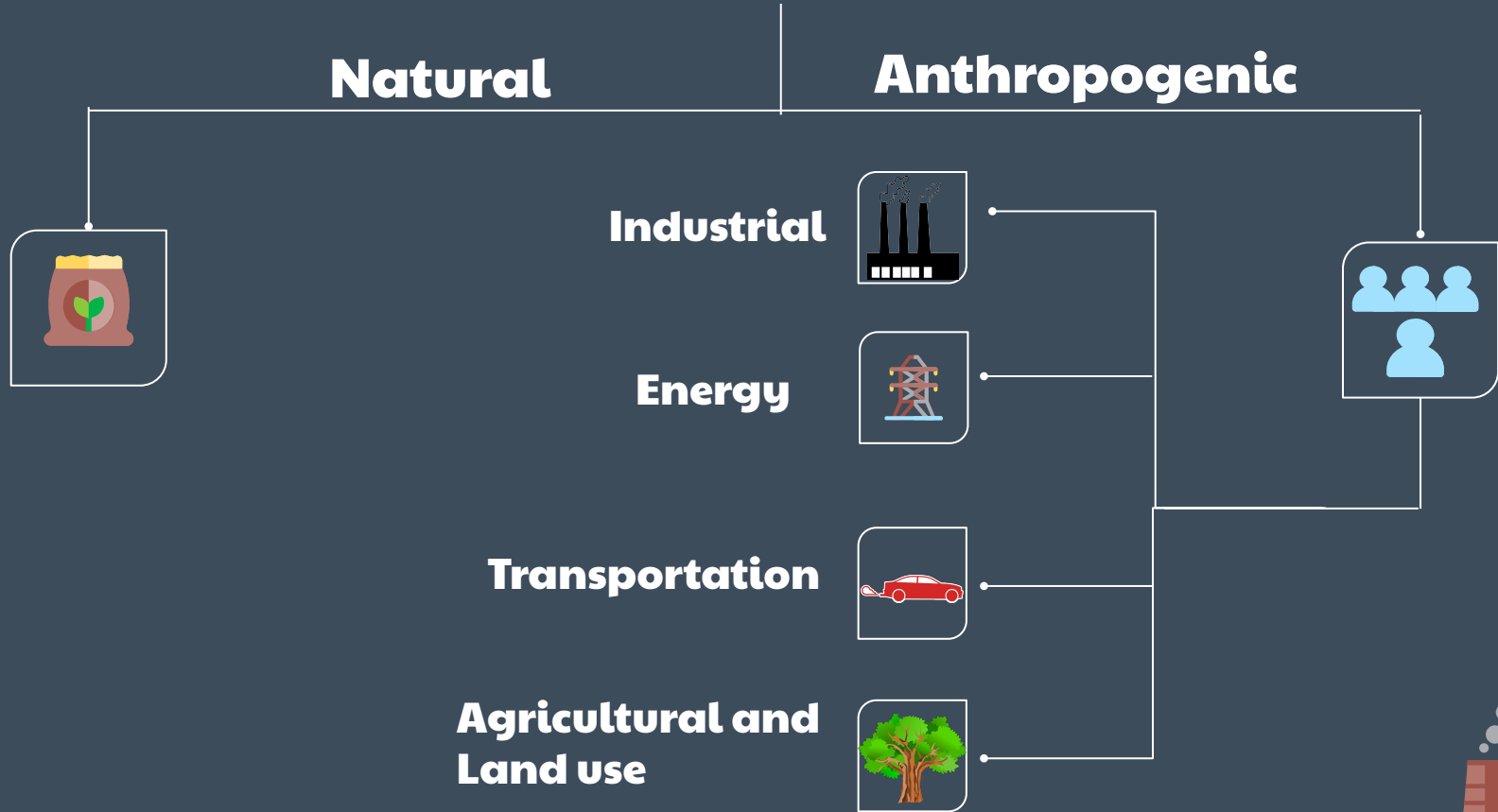


Image Source: IPCC 6th Assessment Report

Emissions all across the globe are increasing



Classification of Carbon Emissions



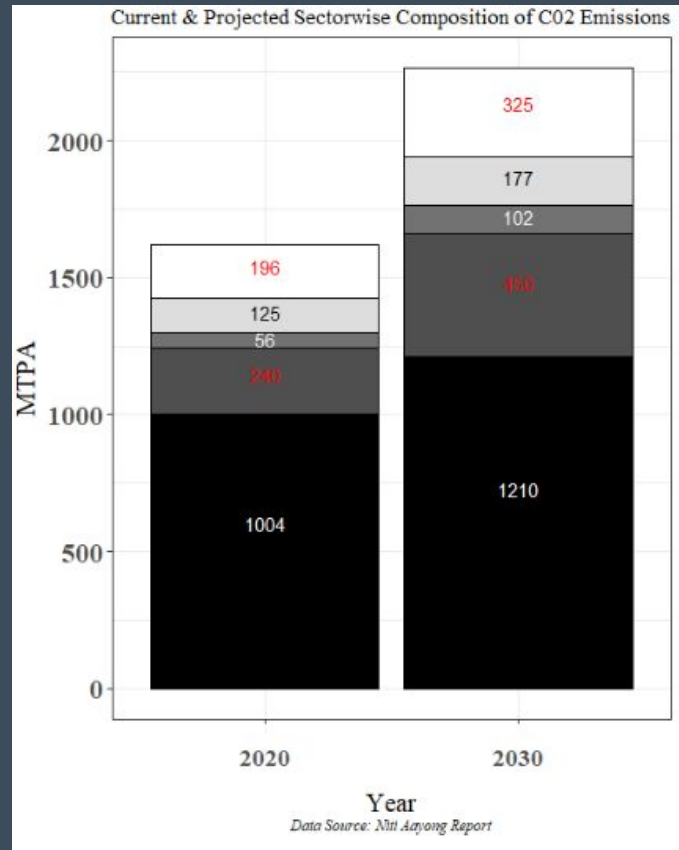
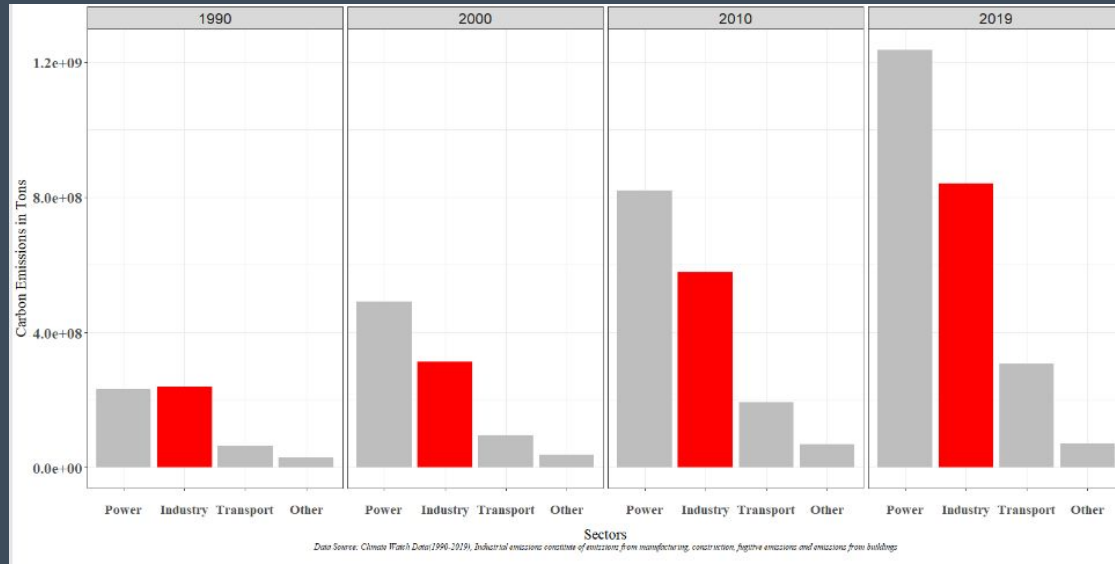
Split of Carbon Emissions



	Sector	Split (%)	Description
	Industrial	23.5	Manufacturing, Construction, Chemicals and Other heavy Industry
	Power	42.2	Burning of fossil fuels to generate electricity
	Transport	16.2	Emissions from automobiles
	Agriculture	5.7	Livestock farming and cultivation

Data Source: IEA, 2021



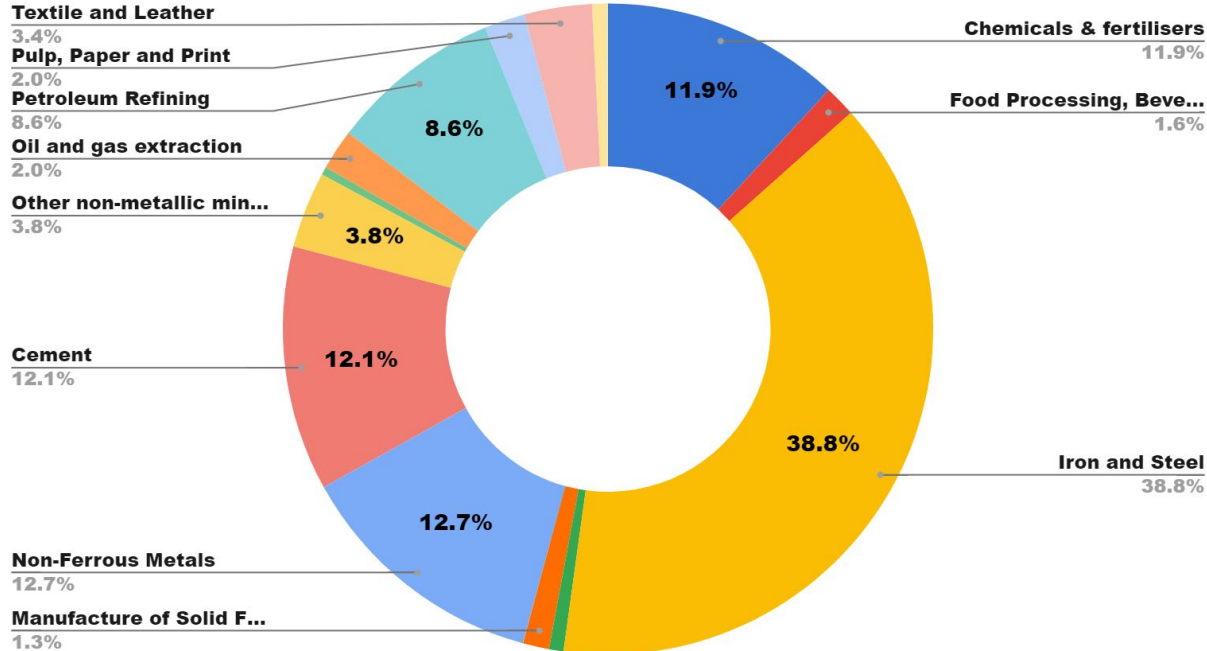


The number of industries are rising, so is industrial emissions; it is projected to rise as well

Industries Responsible For the majority of emissions



HIGHEST EMITTING INDUSTRIES IN 2018





**Majority of these industries are expected
to grow because of Urbanization &
Population Demands**

**Before looking at the possible Solutions,
let us look at the current state of carbon
emissions in India vis-à-vis the World**



Split of Carbon Emissions

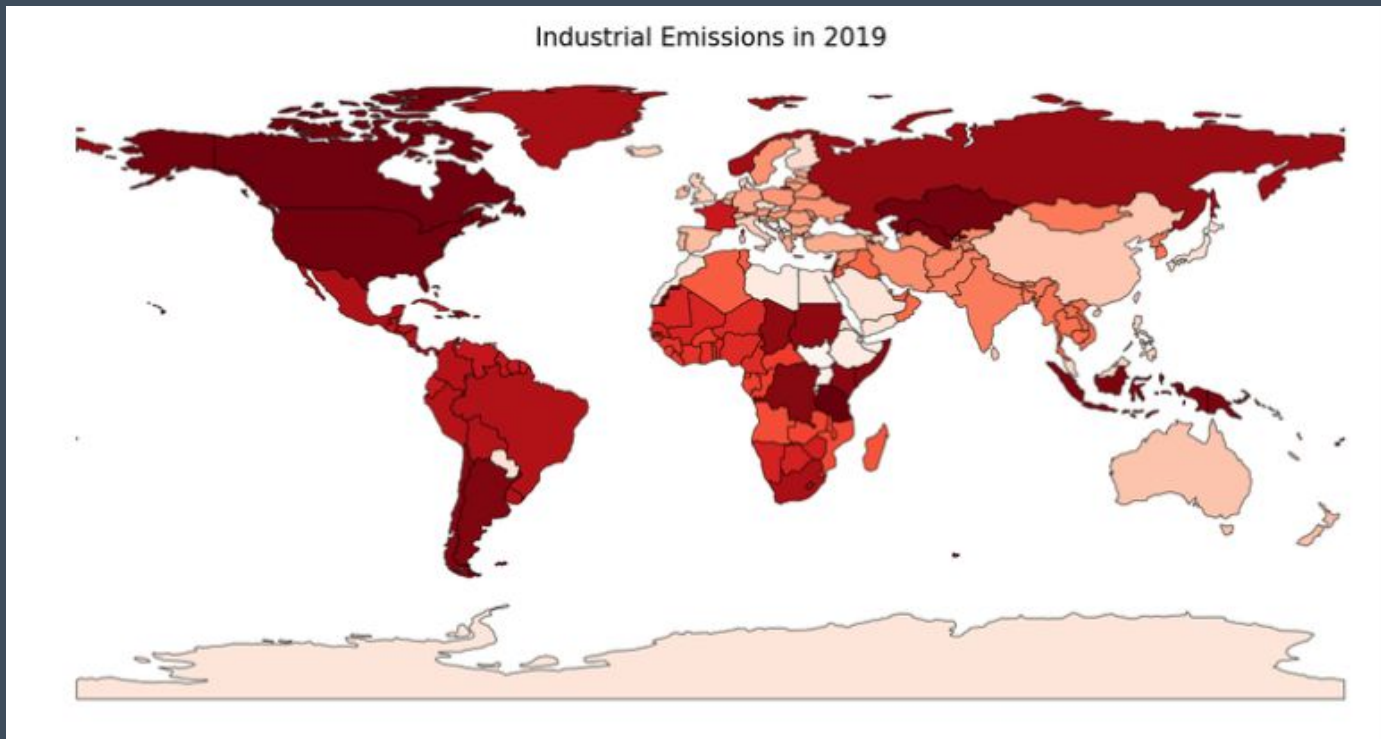


	Sector	India (%)	World (%)	Description
	Industrial	24.5	23.5	Manufacturing, Construction, Chemicals and Other heavy Industry
	Power	46.6	42.2	Burning of fossil fuels to generate electricity
	Transport	7.4	16.2	Emissions from automobiles
	Agriculture	17.2	5.7	Livestock farming and cultivation

Data Source: IEA, 2021

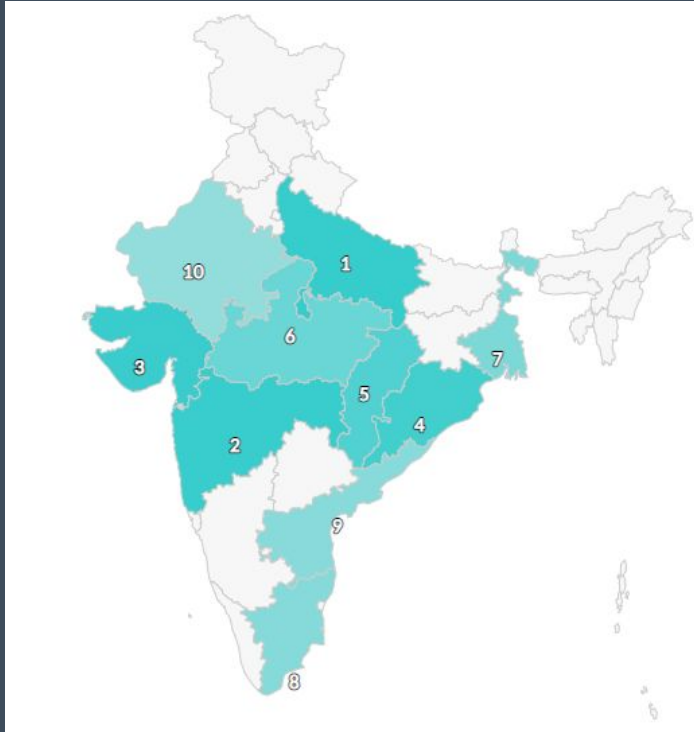


Intensity of Carbon Emissions Worldwide

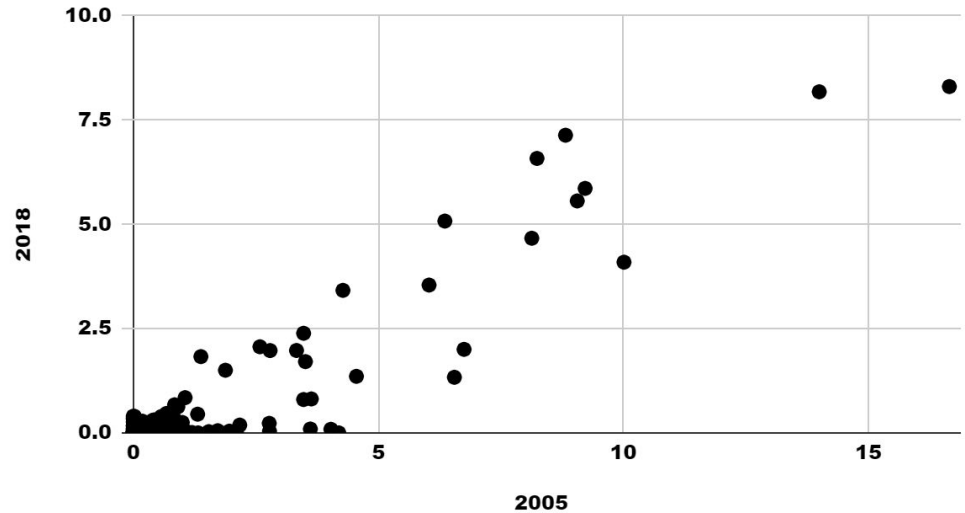




Intensity of Carbon Emissions in India



Statewise Emissions in 2005 compared with 2018





Regressing Growth rate of Emissions On Initial Levels of Emissions in India

$$\left(\frac{\log \frac{Emit_n}{Emit_0}}{T}\right) = \beta_0 + \beta_1 \log(Y_{Emit0}) + \lambda_i + R_j + \epsilon_{ij}$$

In Progress...





Possible Solutions to Rising Industrial Emissions



**Energy
Efficiency**



**Renewable
Energy**



H2 Production



**Sustainable
Agriculture**



CCUS





The Renewable Energy Challenge

	Sector	Could Renewable energy suffice for net zero by 2070?
	Industrial	NO
	Power	YES
	Transport	PERHAPS
	Agriculture	NO



02

Introduction to CCUS

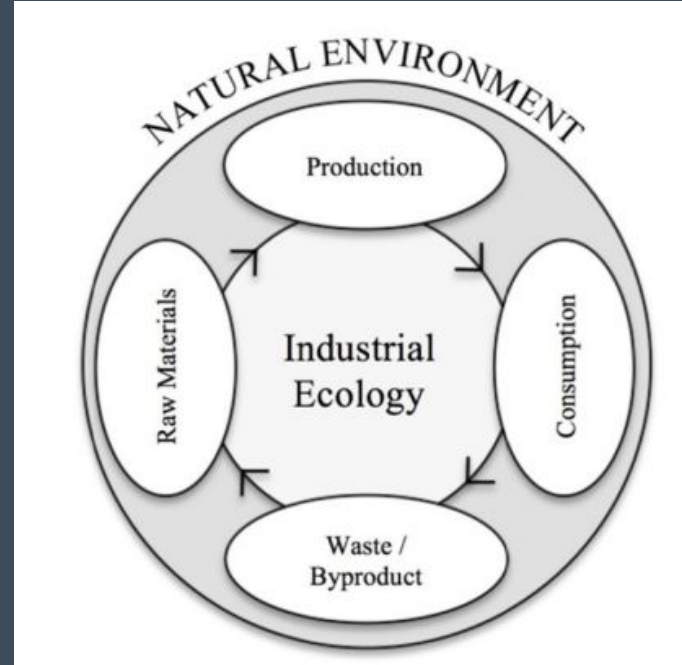
Introduction to the solution, types of technologies involved, and the global status of CCUS operations





Industrial Ecology

Industrial Ecology explores solutions to environmental issues by limiting actions of the industries within the natural environment.



Source : Journal of Sustainable Development







The History of CCUS



1972

**Concept of CCUS
proposed by
JM Smith**



1980

**First commercial
CCUS Project in
Canada**



1997

**Kyoto Protocol is
adapted;
interest in CCUS
rises**



2000s +

**Development of
Large Scale CCUS
Projects Worldwide**



2015

**Paris Agreement
recognizes CCUS
important for
achieving emission
goals**





Types of Carbon Capturing Technologies

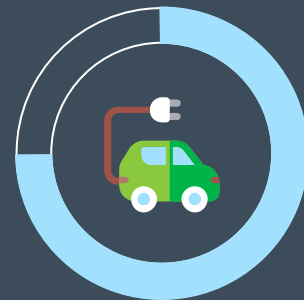


Pre-Combustion

Coal is gasified to produce a synthetic gas containing CO and H₂ gas. CO is later converted into CO₂ by reacting it with water.



Post- Combustion



Oxy-combustion

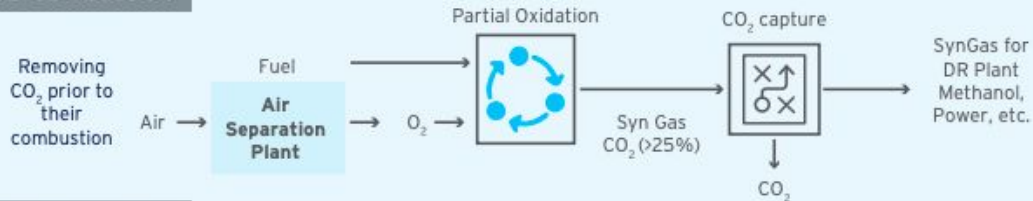
Burning the fuel in oxygen instead of normal air. The waste gas is entirely CO₂ which can be condensed.



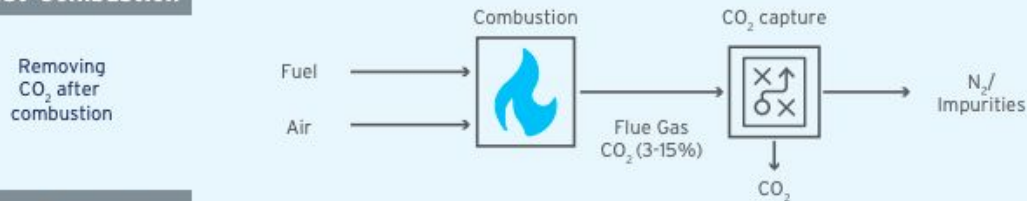


Types of Carbon Capturing Technologies

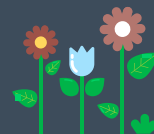
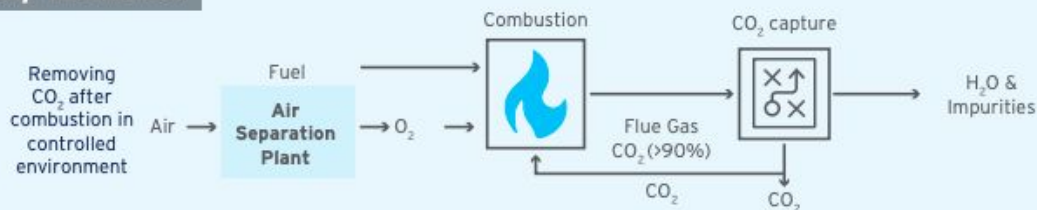
1 Pre-Combustion



2 Post-Combustion

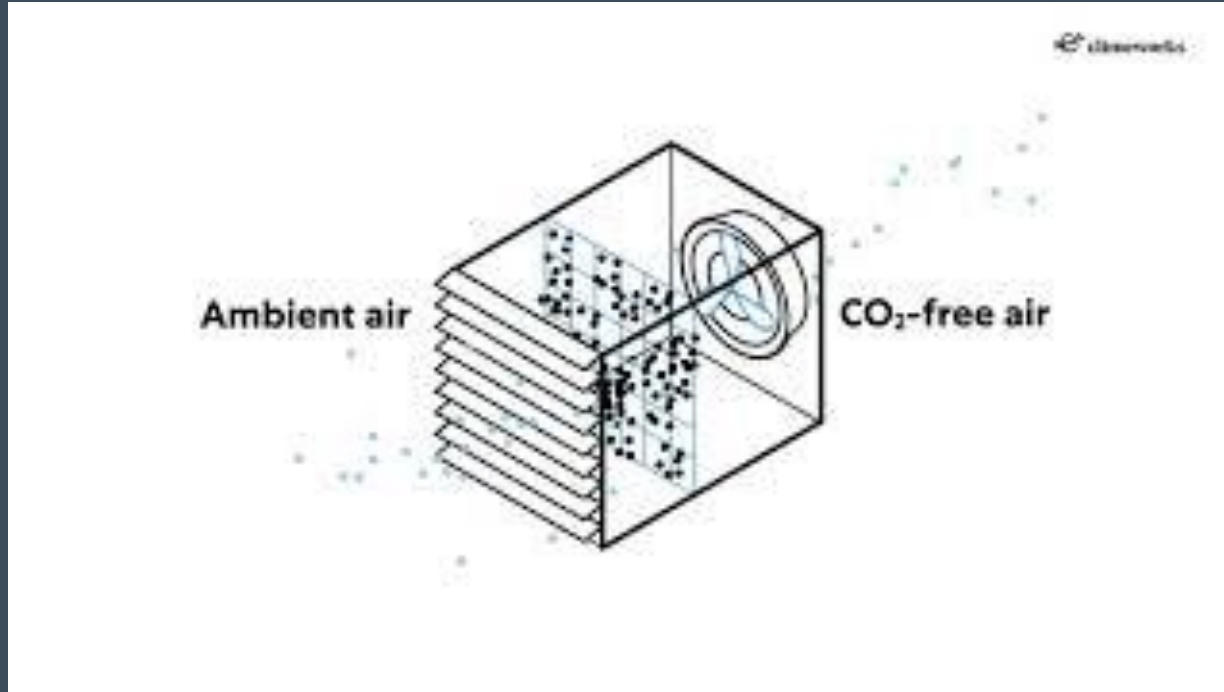


3 Oxy-Combustion





Direct Air Capture (DAC)





What Types of Industries can utilize CO₂?



**Green
Urea**



**Food &
Beverages**



**Oil &
Gas**



**Metal
Fabrication**



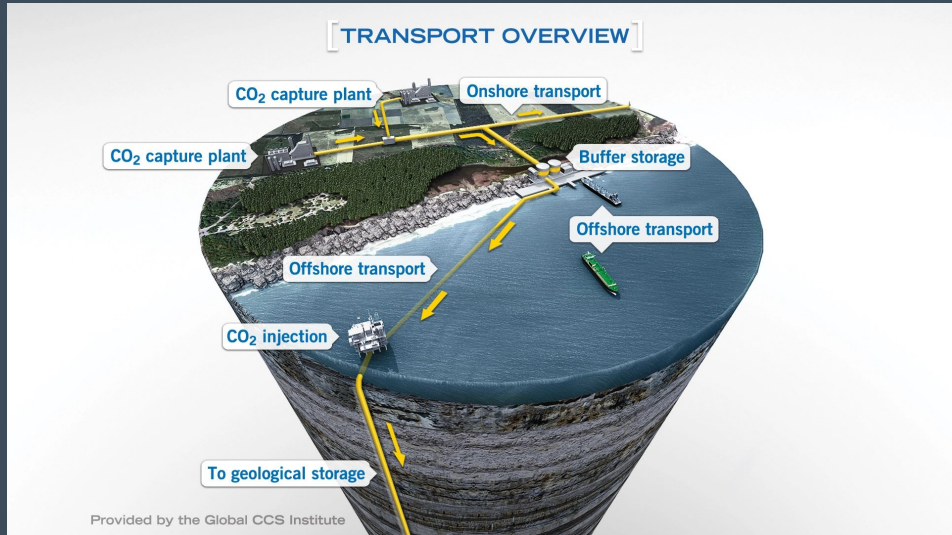
**Cooling
Industry**





Carbon Transportation

- Safe and reliable transportation is necessary due to leakage risks.
- Pipelines, Trucks, Ships, etc. are used for transportation.
- Large-Scale CO₂ transportation similar to LPG transportation which has experienced global supply success.



Source : The Global CCS Institute





CO2 Storage Methods



**Enhanced
Oil Recovery**



**Enhanced Coal
Bed Methane
Recovery**



**Deep Saline
Aquifers**



**Basaltic
Formations**





Enhanced Oil Recovery (EOR)

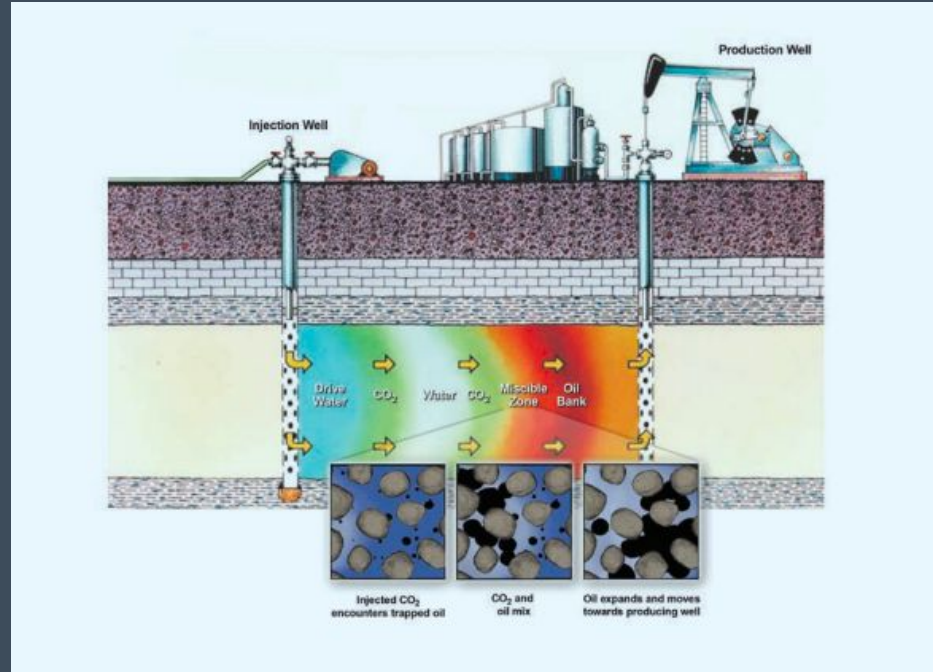


Image Source : NITI Aayog Report, 2022



Enhanced Coal Bed Methane Recovery (ECBMR)

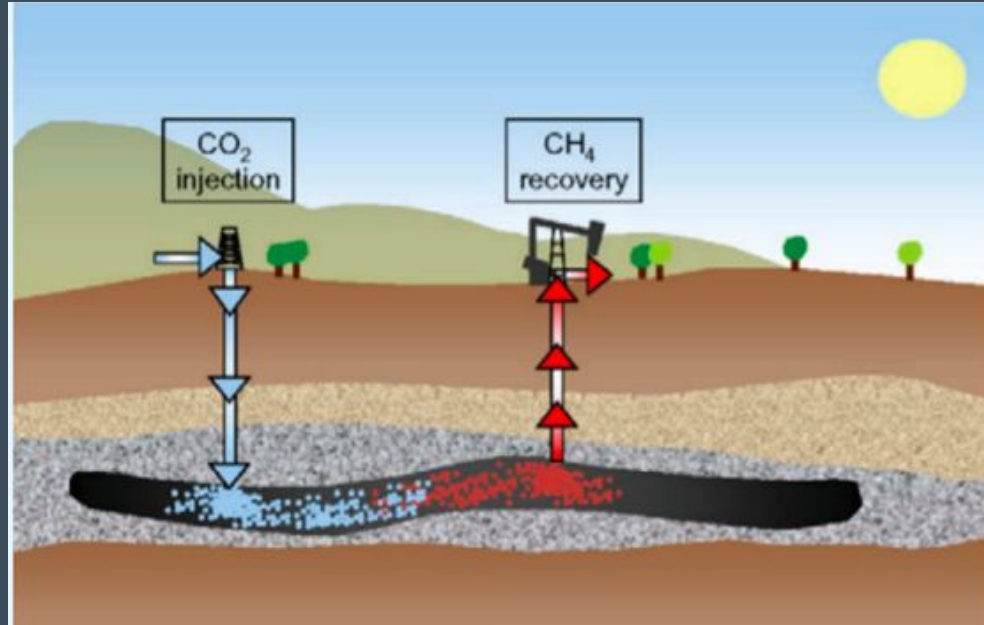


Image Source : NITI Aayog Report, 2022





Deep Saline Aquifers

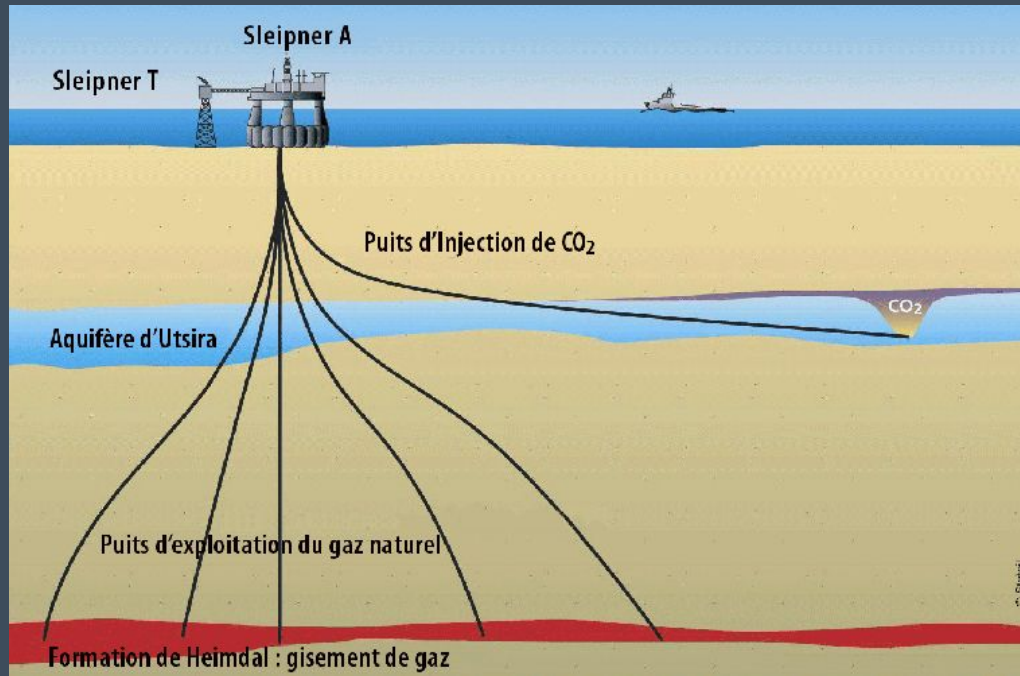
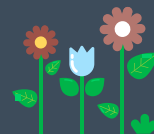


Image Source : Club-CO2 France





Basaltic Formations



Image Source : Club-CO2 France



03

CCUS in India

Evolution of the CCUS discourse in India, Policy Framework (NITI Aayog), Potential Environmental Risks, and Case Studies





Evolution of the CCUS Discourse in India



2007

**Indian CO₂
Sequestration
Applied Research
Network**



2008

**National Action
Plan for Climate
Change (NAPCC)**



2012

Mars

**India's Second
National
Communication
(SNC)**



2015

**Mission
Innovation (MI),
COP-21**

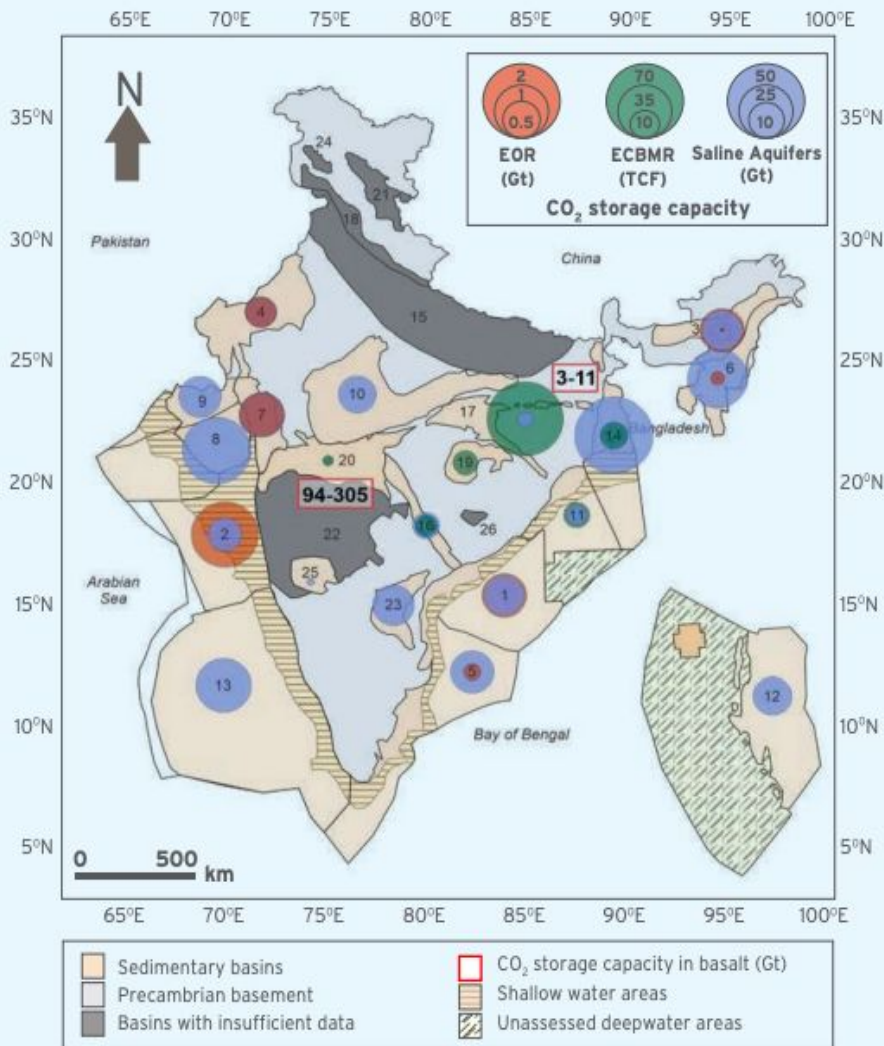




Potential for CO₂ storage in India

Storage Type	Theoretical Storage Capacity(in Mt)
EOR	3.4
ECBMR	3.5–3.7
Deep Saline Aquifers	291
Basalts	91–317





Mapping the potential CO₂ storage sites in India

On the sedimentary basins of India

Map Source : NITI Aayog, 2022

Social Impacts - Future Scope of Work

- The use of land, related displacement of communities and livelihoods leaves the question of procedural and distributive justice in the case of large-scale CO₂ storage sites in India.
- Nielsen.et.al 2022 conduct a systemic meta-narrative literature review to analyse the community acceptance and social risks of major CCUS projects across the project, conceptualising the different kinds of contestations, and recommend improved transparency in such projects.

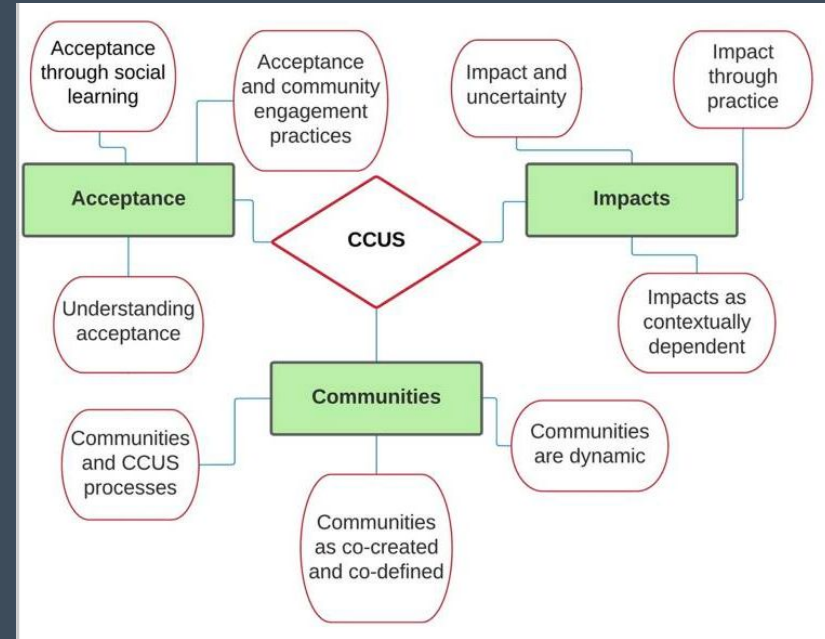


Image Source : Nielsen.et.al., 2022

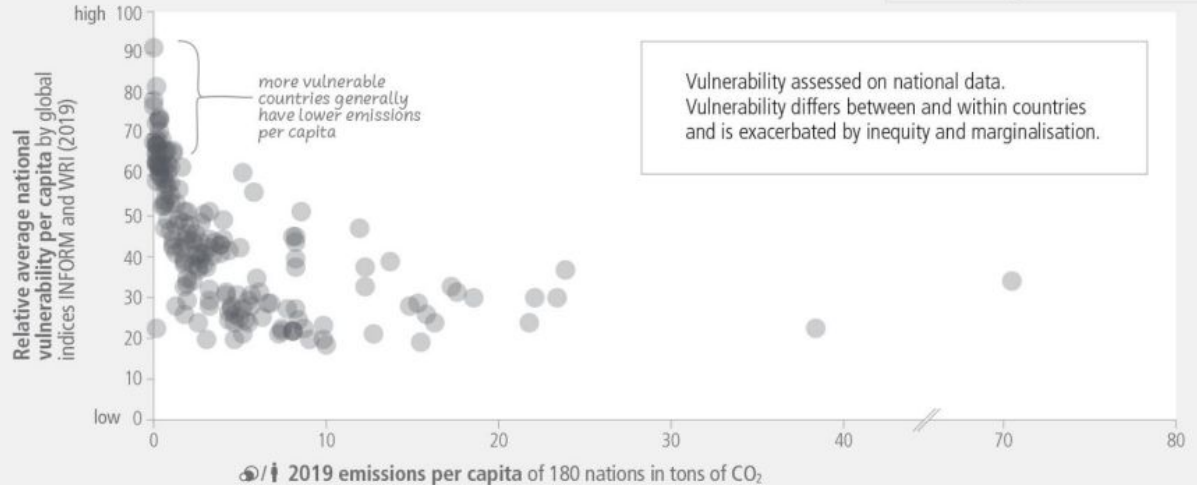


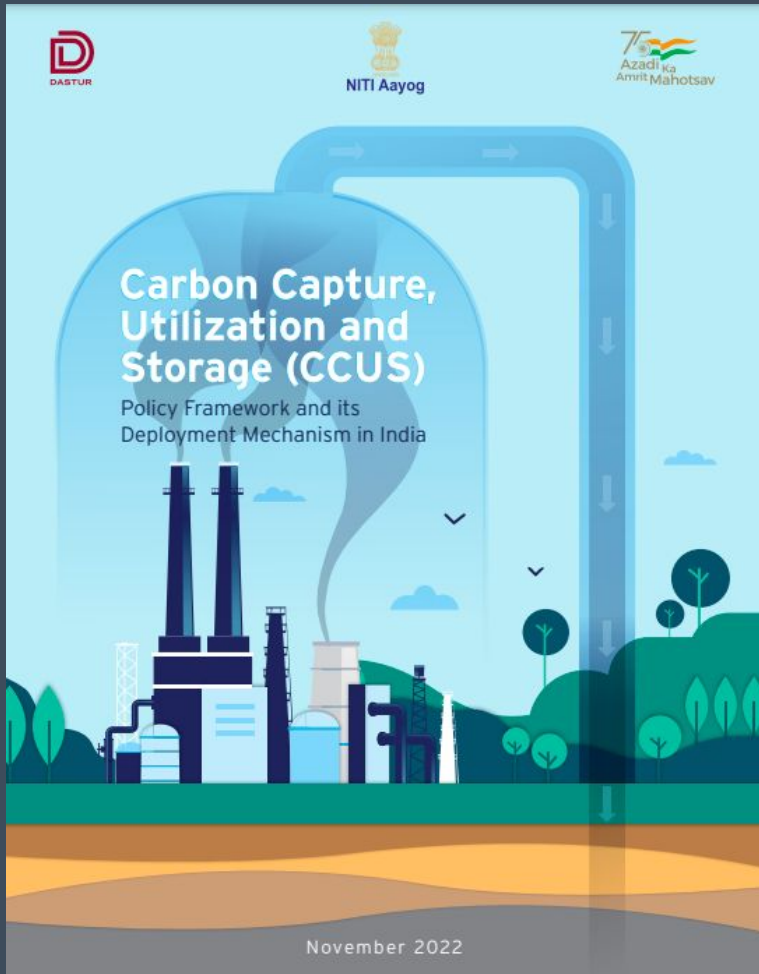
Economic Impacts

- Land availability and displacement
- Will the vulnerable population get adversely affected again?
- Impact on job creation - India had abundance of unskilled workers; CCUS requires skilled.

b) Vulnerability of population & per capita emissions per country in 2019

Dimension of Risk: Vulnerability





NITI Aayog published a report in November, 20200 focusing on the policy frameworks of Carbon Capture, Utilization, and Storage (CCUS) in India.





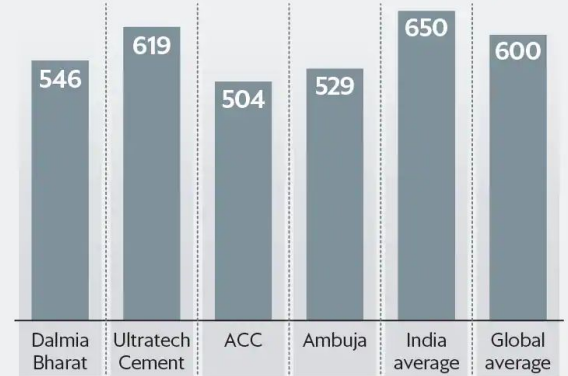
Case Study 1 : Dalmia Cement, TN

- Cement is one of the most hard to abate sectors in the world.
- Building a large scale facility of 500,000 tonne per year carbon capture at its cement plant near Trichy in Tamil Nadu. This is the first large-scale CCUS cement facility in the world.
- The company also plans to achieve "Net-Zero" by 2040.

Say no to carbon

Dalmia Bharat's carbon dioxide emission is lower than the India and global average.

Carbon dioxide emission (kilogram/tonne)



Source: Maybank
Kim Eng Research



Case Study 2 : Koyali Refinery, Gujarat



IndianOil

- The project aims to capture 700,000 tonnes of CO₂ from the refinery's Hydrogen generating units (HGUs).
- Feasibility study completed by Dastur Inc., and funded by USTDA.
- Captured CO₂ will be utilized by ONGC's Gandhar oilfields using EOR and F&B sector.



04

Results and Discussion

The necessity of social and environmental impact assessment, economic cost-benefit analysis, and is it a Yes/No story?



Results and Discussion

- Assess the likely environmental, safety, and health impacts in case of a CO₂ leakage event and evaluate the remedial actions required.
- The economic costs of setting up carbon capture projects is very high with the economic benefits unknown. This makes it hard for a cost-benefit analysis to evaluate the feasibility of most projects.
- CO₂- EOR projects have developed technologies for many years and thus affordable than other types of projects.
- Social and ecological impacts need to be considered while choosing the storage sites as community acceptance and awareness is crucial.
- NITI Aayog report suggests that carbon markets would incentivise CCUS projects and recommends establishing a Carbon Capture Finance Corporation (CCFC) to support CCUS projects through their lifecycle.